File permissions in Linux

Project description

Security Professional in the research team and examining the file permissions determining who will have access to them and what kind of permissions and removing the unauthorized permissions.

Check file and directory details

```
researcher2@494825172b37:~$ ls
projects
researcher2@494825172b37:~$ cd projects
researcher2@494825172b37:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:31 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:58 ..
-rw--w---- 1 researcher2 research_team   46 Oct 23 13:31 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 23 13:31 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Oct 23 13:31 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Oct 23 13:31 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_t.txt
researcher2@494825172b37:~/projects$ chmod o-w project_k.txt
researcher2@494825172b37:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:31 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:58 ..
-rw--w---- 1 researcher2 research_team   46 Oct 23 13:31 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 23 13:31 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Oct 23 13:31 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_t.txt
```

Describe the permissions string

In permission string we use d for directory and - for the file
Eg: drwxrwxrwx and -rwxrwxrwx

Change file permissions

Chmod is used to change the permissions of the different file or directories

```
researcher2@494825172b37:~/projects$ chmod g-rw project_m.txt
researcher2@494825172b37:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:31 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:58 ..
-rw--w---- 1 researcher2 research_team   46 Oct 23 13:31 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 23 13:31 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_k.txt
-rw------- 1 researcher2 research_team   46 Oct 23 13:31 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_t.txt
researcher2@494825172b37:~/projects$ chmod g-w,u-w project_x.txt
chmod: cannot access 'project_x.txt': No such file or directory
researcher2@494825172b37:~/projects$ chmod g-w,u-w .project_x.txt
researcher2@494825172b37:~/projects$ chmod u=r,g=r .project_x.txt
researcher2@494825172b37:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:31 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:58 ..
-r--r----- 1 researcher2 research_team   46 Oct 23 13:31 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 23 13:31 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_k.txt
-rw------- 1 researcher2 research_team   46 Oct 23 13:31 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_t.txt
```

Change file permissions on a hidden file

```
researcher2@494825172b37:~/projects$ chmod g-rw project_m.txt
researcher2@494825172b37:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:31 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:58 ..
-rw--w---- 1 researcher2 research_team   46 Oct 23 13:31 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 23 13:31 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_k.txt
-rw------- 1 researcher2 research_team   46 Oct 23 13:31 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_t.txt
researcher2@494825172b37:~/projects$ chmod g-w,u-w project_x.txt
chmod: cannot access 'project_x.txt': No such file or directory
researcher2@494825172b37:~/projects$ chmod g-w,u-w .project_x.txt
researcher2@494825172b37:~/projects$ chmod u=r,g=r .project_x.txt
researcher2@494825172b37:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:31 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:58 ..
-r--r----- 1 researcher2 research_team   46 Oct 23 13:31 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 23 13:31 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_k.txt
-rw------- 1 researcher2 research_team   46 Oct 23 13:31 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_t.txt
```

Change directory permissions

```
researcher2@494825172b37:~/projects$ chmod u-x drafts
researcher2@494825172b37:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:31 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:58 ..
-r--r----- 1 researcher2 research_team   46 Oct 23 13:31 .project_x.txt
drw---x--- 2 researcher2 research_team 4096 Oct 23 13:31 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_k.txt
-rw------- 1 researcher2 research_team   46 Oct 23 13:31 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_t.txt
researcher2@494825172b37:~/projects$ chmod g-x drafts
researcher2@494825172b37:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:31 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 23 13:58 ..
-r--r----- 1 researcher2 research_team   46 Oct 23 13:31 .project_x.txt
drw------- 2 researcher2 research_team 4096 Oct 23 13:31 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_k.txt
-rw------- 1 researcher2 research_team   46 Oct 23 13:31 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 23 13:31 project_t.txt
researcher2@494825172b37:~/projects$ chmod u+x drafts
researcher2@494825172b37:~/projects$
```

Summary

After analyzing the permissions of the different files and directories there were some unauthorized permissions and for better security i removed those permissions